# HealthScope: A Public Health Analytics and Diabetes Risk Classification Platform Using Machine Learning

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## ABSTRACT

Health surveys collect information about people's health, daily habits and general well-being. Looking through such large datasets manually can make it difficult to notice useful patterns. This paper presents HealthScope, a web-based application that brings data analysis and diabetes-status prediction together in one place. The system works with 253,680 records and 21 input features, and uses Logistic Regression to classify a record as No Diabetes, Prediabetes or Diabetes. Users can enter health-related details through a Streamlit form, view the predicted category and explore charts that summarize the dataset. The model achieved 65.41% accuracy, 84.89% weighted precision, 65.41% weighted recall and 72.54% weighted F1-score when evaluated on the full dataset. The results also show that Prediabetes is difficult for the model to identify. Since the evaluation used the same dataset on which the model was trained, these figures should not be treated as performance on unseen data. HealthScope is intended as an educational and analytical project, not as a tool for medical diagnosis.

Index Terms— Public health analytics, diabetes risk classification, Logistic Regression, exploratory data analysis, machine learning, Streamlit, health indicators.

## I. INTRODUCTION

Diabetes is a major public-health concern, and understanding the factors associated with it can help people and researchers make better use of health information. Survey datasets often contain details about blood pressure, cholesterol, physical activity, diet, age and other health-related factors. When these records are considered together, they can reveal patterns that may be difficult to spot by examining individual entries.

HealthScope was developed to make this kind of analysis easier to explore. It brings data preparation, visual analysis and machine-learning-based classification into a single web application. The project follows five main stages: Collect, Clean, Analyze, Predict and Visualize. A user can enter 21 health and lifestyle details and receive one of three predicted categories: No Diabetes, Prediabetes or Diabetes. The dashboard is designed to make the process understandable, while keeping clear that a model prediction is not a medical diagnosis.

The project brings together three practical features: visual summaries of the health dataset, an input form for making a prediction, and a model-performance section that helps users understand how well the classifier works. It also converts the age entered by the user into the category expected by the dataset and applies the same standardization process used when training the model.

## II. RELATED WORK

Health-related datasets have been used in many studies to understand disease patterns and support public-health research. One important source is the Behavioral Risk Factor Surveillance System (BRFSS), which collects information about health conditions and lifestyle behaviours from adults in the United States [1]. The Diabetes Health Indicators dataset used in this project is based on BRFSS information and provides a way to study how different health indicators relate to diabetes status [2].

Logistic Regression is a widely used method for classification. It estimates the probability of an input belonging to each available class and is often chosen as a useful baseline because its behaviour can be examined [3]. When input variables use different scales, standardization can also help the model process them consistently. Scikit-learn provides the StandardScaler utility for this purpose [4].

HealthScope uses these ideas in a single application by combining exploratory data analysis, a Logistic Regression classifier and a simple dashboard. The current project focuses on Logistic Regression. Comparing it with methods such as Decision Tree, Random Forest and K-Nearest Neighbours is considered future work, rather than a result already completed.

## III. DATASET AND PREPROCESSING

The application reads the file healthscope_data.csv. It contains 253,680 records and 22 columns: 21 input features and one target column, Diabetes_012. The dataset is based on the Diabetes Health Indicators data derived from the CDC BRFSS survey [1], [2]. The target column has three possible values: 0 for No Diabetes, 1 for Prediabetes and 2 for Diabetes.

The three groups are not represented equally in the dataset. There are 213,703 No Diabetes records (84.24%), 4,631 Prediabetes records (1.83%) and 35,346 Diabetes records (13.93%). The project documentation reports that there are no missing values and 23,899 duplicate rows (9.42%). These duplicate rows were retained in the dataset used by the dashboard.

The input features cover several aspects of a person's health and lifestyle. They include indicators such as high blood pressure, high cholesterol, smoking, stroke history, physical activity, fruit and vegetable consumption, and access to healthcare. Other fields describe general health, age, education and income, while BMI and the number of physically or mentally unhealthy days are recorded as numerical values. The form asks users to enter their age in years and converts it into one of the 13 age groups used in the dataset.

For a prediction, the application arranges the entered values into a single-row DataFrame. It keeps the feature names and order consistent with the training data, then uses the saved StandardScaler to transform the values. This is important because the model expects inputs to be processed in the same way as the data it was trained on.

## IV. PROPOSED METHODOLOGY

HealthScope uses a Logistic Regression model to classify each input into one of the three diabetes-status categories. The model is implemented with Scikit-learn [3]. According to the project configuration, it uses L2 regularization, C = 1.0, the lbfgs solver, a maximum of 1,000 iterations and random_state = 42. The training process reportedly converged in 11 iterations. Before classification, the 21 input features are standardized using the saved scaler.

When a user submits the form, the application follows these steps:

- Load the trained Logistic Regression model and StandardScaler with Joblib.
- Collect the 21 health, lifestyle and demographic details entered in the Streamlit form.
- Convert the age entered in years into the matching age category (1 to 13).
- Arrange the values in the same feature order used during model training.
- Standardize the input values with the saved StandardScaler.

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• Use model.predict() to classify the record as No Diabetes, Prediabetes or Diabetes.

• Show the predicted category in the result card along with a clear educational disclaimer.

The application is organised into connected parts for loading the data, preparing the input, running the machine-learning model, displaying the prediction and presenting analytics. These parts run within the same Streamlit application. The dataset, trained model and scaler are loaded when the application starts; the current version does not use a separate REST API backend.

## V. SYSTEM IMPLEMENTATION

HealthScope's interface is built with Streamlit. It provides the page layout, tabs, input fields, metric cards, charts and prediction display. Custom CSS is used to give the dashboard a consistent appearance. The supporting Python code handles data loading, preprocessing, prediction and evaluation. Pandas is used to work with the dataset, NumPy supports numerical operations, and Scikit-learn provides the classifier, scaler and evaluation metrics. Matplotlib is used to display the confusion matrix, while Joblib saves and loads the trained model and scaler.

The dashboard is divided into four main sections. Health Prediction contains the form where users enter clinical, lifestyle and personal details. Public Health Analytics presents summaries and charts for diabetes status, BMI, blood pressure, physical activity and general health. Model Performance displays the evaluation measures and confusion matrix. The About HealthScope section explains the purpose and workflow of the application.

To run the application locally, Python and the required packages must be installed. The dashboard can then be started from the project folder with the command: python -m streamlit run app.py. The current version is intended for local use, although cloud deployment could be considered as a future improvement.

## VI. EXPERIMENTAL RESULTS AND DISCUSSION

The class-wise results show that performance differs considerably between the three categories. For No Diabetes, precision is 94.89%, recall is 67.21% and the F1-score is 78.69% (213,703 records). For Prediabetes, the corresponding values are 3.16%, 29.19% and 5.71% (4,631 records). For Diabetes, precision is 35.15%, recall is 59.22% and the F1-score is 44.11% (35,346 records).

The model gives its strongest results for the No Diabetes group, which also makes up most of the dataset. It correctly identifies 20,931 of the 35,346 Diabetes records, giving a recall of 59.22%. Prediabetes is much harder to identify: the model recognises 1,352 of the 4,631 records, or 29.19%. Its precision for this class is only 3.16%, meaning that many records predicted as Prediabetes belong to another class. This is an important limitation and shows why accuracy alone does not tell the whole story, especially when the classes are imbalanced.

The exploratory analysis also shows differences across the three groups. The average BMI is 27.74 for No Diabetes, 30.72 for Prediabetes and 31.94 for Diabetes. The proportion of records reporting high blood pressure is 37.1%, 62.9% and 75.3%, respectively. Physical activity shows the opposite pattern: it is reported by 77.9% of the No Diabetes group, 67.8% of the Prediabetes group and 63.1% of the Diabetes group. These summaries help describe patterns in the dataset, but they do not show that any one factor causes diabetes.

The project documentation also reports positive correlations between Diabetes_012 and general health (0.30), high blood pressure (0.27), BMI (0.22), difficulty walking (0.22), high cholesterol (0.21) and age (0.19). These values describe relationships observed in the dataset; they should not be interpreted as proof of cause and effect. Another important point is that the model was evaluated on the full dataset, including the records used for training. The reported metrics are therefore not a measure of how accurately the model will perform on completely new records.

## VII. LIMITATIONS AND FUTURE WORK

There are a few limitations to keep in mind when interpreting HealthScope. The dataset is imbalanced, with Prediabetes making up less than two per cent of the records, and the model has difficulty identifying this group. The evaluation was carried out on the complete dataset rather than on a separate test set, so the results do not establish how well the model will work with unseen data. The dataset is based on survey responses and does not include laboratory test results such as blood-glucose measurements. In addition, the current implementation uses Logistic Regression only; other algorithms have not yet been tested in the project.

Several improvements can be explored in future versions. The model could be compared with Decision Tree, Random Forest and K-Nearest Neighbours. Class weights or resampling methods may help address the imbalance between the target categories. A more reliable evaluation should use a separate test set and cross-validation. Other possible additions include displaying class probabilities, explaining individual predictions, allowing users to download reports, adding filters to the dashboard, improving loading speed and deploying the application online. Any future version should continue to make clear that its output is not a medical diagnosis.

## VIII. CONCLUSION

HealthScope brings together public-health data analysis and diabetes-status classification in one interactive Streamlit application. It works with 253,680 survey records and 21 input features, and uses Logistic Regression to classify profiles as No Diabetes, Prediabetes or Diabetes. Alongside the prediction form, the dashboard provides charts and model-performance information that help users explore the dataset. The model achieved 65.41% accuracy and a weighted F1-score of 72.54% when evaluated on the full dataset. However, the results also reveal a clear weakness in identifying Prediabetes. Further work, especially testing on unseen records and addressing class imbalance, is needed before the model's performance can be judged for practical use. HealthScope should be viewed as an academic project for analysis and learning, not as a substitute for professional medical advice.

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